

Internship Internal Project Business Cases

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Student Handout for Project Selection, Planning, and Execution

Purpose: This document presents three feasible internship project ideas that can be completed within a two-month internship period. Each business case is designed to help student interns understand the business problem, proposed solution, expected scope, deliverables, and implementation plan.

Project Option	Business Value	Complexity	Recommended Team Size
Option 1: Industry Session & Event Management System with Alumni & Internship Vacancy Portal	Very High - supports IPT, guest lectures, workshops, attendance, feedback, and certificates and supports internship placement, alumni engagement, company vacancies, and student applications	Medium	9-10 interns
Option 2: Student Mentoring Management System	High - supports mentor allocation, meeting tracking, student progress, and issue escalation	Medium	4-5 interns

Common Expectations for All Student Teams

- Duration: 2.5 months / 10 weeks.
- Expected output: Working MVP, source code, database design, UI screens, test evidence, user manual, deployment guide, and final presentation.
- Recommended development approach: Agile / Scrum with weekly progress reviews.
- Minimum quality expectation: Authentication, role-based access, data validation, responsive UI, dashboard, exportable reports, and basic security controls.
- Suggested technology stack: React / Angular / Laravel / Node.js / Spring Boot / .NET with MySQL, PostgreSQL, MongoDB, or Firebase. The final stack may be selected based on student capability.
- In addition to the system development, each intern team is expected to prepare a basic digital marketing plan for the proposed solution. The plan should identify the target users, key value proposition, suitable digital channels, sample promotional content, and a simple launch strategy. Students should also propose how the platform can be introduced to its intended users and how user engagement can be improved after deployment. This will help students understand not only software development, but also product positioning, user adoption, and business value creation.
- Students are encouraged to add any additional features based on the capability and technical strength of their team. Any practical, user-friendly, and innovative feature that improves the overall value of the system will be considered positively. If the team can introduce AI-based features such as smart recommendations, automated reminders, chatbot support, predictive analytics, intelligent search, or AI-assisted report generation, it will add further value to the project. However, all additional features should be realistic, relevant to the project scope, and achievable within the 2-month project duration.

Option 1: Industry Session & Event Management System with Alumni & Internship Vacancy Portal

1. Business Problem

The Faculty conducts many industry sessions, IPT programmes, guest lectures, workshops, and training events. However, planning sessions, maintaining speaker details, managing attendance, collecting feedback, and generating participation records are often handled through separate documents and manual processes. This creates duplication, delays, and difficulty when preparing reports.

At the same time, internship opportunities are often shared through emails, personal contacts, alumni networks, and company representatives. Without a centralized portal, it is difficult to maintain a live vacancy list, track student applications, verify alumni-provided opportunities, and monitor placement progress.

Because industry sessions and internship placement involve the same stakeholders (students, alumni, industry partners, and the Industry Collaboration Unit), maintaining two separate manual processes leads to duplicated company and contact records, fragmented student engagement data, and no single view of the Faculty's overall industry engagement.

2. Proposed Solution

Develop a single centralized web-based platform that manages both industry engagement events and internship placement. The system will support event scheduling, speaker management, student registration, QR-based attendance, feedback collection, certificate eligibility, and management dashboards. On the same platform, alumni and industry contacts can submit internship vacancies, faculty coordinators can verify and publish them, and students can view and apply for suitable opportunities while the system tracks application and placement status.

A shared database of companies, alumni, speakers, and students connects both modules, so a company that delivers a guest lecture can also post vacancies, and student participation in events and internship applications can be viewed in one profile.

3. Business Objectives

- Centralize all industry session, event, alumni, and internship vacancy information in one platform.
 - Reduce manual attendance, feedback, and vacancy handling.
 - Maintain a proper history of speakers, sessions, companies, alumni, and student participation.
 - Support certificate generation based on attendance and feedback completion.
 - Strengthen alumni contribution to internship placement.
 - Maintain a centralized and verified internship vacancy list and enable students to apply through one platform.
 - Track applications, selections, and placement status.
 - Provide quick, combined reports on events, engagement, vacancies, and placement for management review and faculty planning..

4. Key Stakeholders

Industry Collaboration Unit / Faculty coordinator, Internship coordinators, Students, Alumni, Industry speakers and partners, Academic staff, Secretary / administrative staff, Faculty management..

5. MVP Scope for 2 Months

Included in MVP	Not Included in MVP / Future Scope
Core web application with login and user roles (shared across both modules)	Advanced AI analytics

Event & session management module (Module A) with main data entry and management screens	Full mobile app unless the team has sufficient capacity
Alumni & internship vacancy module (Module B) with vacancy submission, verification, and applications	Integration with university ERP/LMS
Combined dashboard and basic reports	Payment gateway or complex third-party integrations
Basic notifications by email or system alert	Enterprise-level security audit
Testing, documentation, and deployment guide	Automated CV matching and interview scheduling (future scope)

6. Key Functional Requirements

Shared Core

- Admin login and role-based access (Admin, Coordinator, Student, Alumni/Company, Management)
- Company / alumni / speaker profile management (single shared directory)
- Student profile management with CV upload
- Dashboard: total sessions, attendance, feedback rate, active vacancies, applications, placed students, top companies
- Export reports to Excel/PDF (by company, degree, batch, event, and placement status)

Module A – Industry Session & Event Management

- Create and manage events/sessions
- Student registration for events
- QR-based attendance marking
- Feedback form after each session
- Certificate eligibility status

Module B – Alumni & Internship Vacancy Portal

- Alumni/company registration
- Vacancy submission form
- Admin verification and approval of vacancies
- Vacancy search and filtering
- Student application tracking
- Status updates: Applied, Shortlisted, Selected, Rejected, Placed

7. Non-Functional Requirements

- Responsive user interface suitable for laptop and mobile browser use.
 - Secure login and role-based access control.
 - Input validation and proper error messages.
 - Basic audit trail for important updates.
 - Database backup and restore instructions.
 - Usability-focused screens with simple navigation.

8. Sample User Stories

User Role	User Story
Admin	As an admin, I want to create and manage user accounts so that system access is controlled.
Coordinator	As a coordinator, I want to view a combined dashboard of events and vacancies so that I can monitor progress quickly.
Student	As a student, I want to register for industry sessions and apply for internship vacancies from one platform.
Alumni / Company	As an alumnus or company representative, I want to submit internship vacancies so that students can apply for verified opportunities.
Management	As faculty management, I want to download reports so that decisions can be made using updated data.

9. Expected Deliverables

- Project charter and requirement specification.
 - UI wireframes and final screen designs.
 - Database ER diagram and schema.
 - Working MVP with source code.
 - Test plan, test cases, defect log, and UAT evidence.
 - User manual and deployment guide.
 - Final presentation and project demonstration.

10. Suggested Future Enhancements

- Automated certificate generation
 - Email/SMS reminders and notifications
 - Integration with LMS
 - Mobile app for attendance scanning
 - Speaker rating and recommendation analytics
 - Automated CV matching
 - Company interview scheduling
 - Alumni engagement analytics
 - LinkedIn profile integration

Recommended 8-Week Implementation Plan

Week	Activities	Expected Output
Week 1	Requirement gathering for both modules, stakeholder discussion, user stories, high-level scope confirmation	Approved project brief and product backlog
Week 2	UI wireframes, database design (shared company/alumni/student entities), architecture design, development environment setup	Wireframes, ER diagram, architecture diagram
Week 3	Authentication, role management, shared core master data screens (companies, alumni, speakers, students)	Login, user roles, main setup screens
Week 4	Module A development: event creation, student registration, QR attendance, feedback, certificate eligibility	Event management features completed
Week 5	Module B development: vacancy submission, admin verification, vacancy search, student applications and status tracking; combined dashboard, reporting, export options, notifications	Vacancy portal features, management dashboard and reports
Week 6	System testing across both modules, bug fixing, usability improvements	Test evidence and improved build
Week 7	UAT with selected users, refinements, documentation	UAT feedback and final adjustments
Week 8	Final deployment preparation, handover, presentation, demonstration	Final MVP, documentation, demo presentation

Recommended Intern Team Structure

Week	Activities	Expected Output
Week 1	Requirement gathering for both modules, stakeholder discussion, user stories, high-level scope confirmation	Approved project brief and product backlog
Week 2	UI wireframes, database design (shared company/alumni/student entities), architecture design, development environment setup	Wireframes, ER diagram, architecture diagram
Week 3	Authentication, role management, shared core master data screens (companies, alumni, speakers, students)	Login, user roles, main setup screens
Week 4	Module A development: event creation, student registration, QR attendance, feedback, certificate eligibility	Event management features completed
Week 5	Module B development: vacancy submission, admin verification, vacancy search, student applications and status tracking; combined dashboard, reporting, export options, notifications	Vacancy portal features, management dashboard and reports
Week 6	System testing across both modules, bug fixing, usability improvements	Test evidence and improved build

Week 7	UAT with selected users, refinements, documentation	UAT feedback and final adjustments
Week 8	Final deployment preparation, handover, presentation, demonstration	Final MVP, documentation, demo presentation

Project Operation Plan

The internship internal project will be conducted over a period of two months. The project team is expected to work approximately 8 to 9 hours per day, similar to a professional working environment. Team members should maintain discipline, punctuality, accountability, and proper communication throughout the project period.

At the initial stage, students must finalize their team members, team lead, project idea, and proposed roles. Once finalized, the team should formally inform the Internship Internal Project Evaluation Team via email. The email should be addressed to Ms. Pavithra, Ms. Nimesha, and Mr. Savindu, while copying Mr. Poorna, Secretary of the Industry Collaboration Unit. The email should include the project title, team member details, assigned roles, contact details, and the selected project idea.

Each team must maintain a daily project diary to record daily work progress, tasks completed, issues faced, decisions made, and next-day action items. This diary will be considered as part of the project monitoring process and may be reviewed during evaluation meetings.

The team is required to provide a weekly status update to Ms. Nimesha and Mr. Savindu. The weekly update should include completed tasks, ongoing tasks, pending activities, risks/issues, support required, and the planned work for the following week. Ms. Pavithra will join the first meeting of each month to review the overall progress, provide guidance, and confirm whether the project is moving according to the expected direction.

The team must actively participate in all scheduled review meetings, maintain proper documentation, and ensure that the project deliverables are completed within the given two-month timeline. Any delays, scope changes, technical difficulties, or team-related issues should be communicated immediately to the evaluation team.

Option 2: Student Mentoring Management System

1. Business Problem

Mentoring is important for student guidance, academic progress, employability readiness, and early identification of student issues. However, mentor allocation, meeting schedules, mentoring notes, and follow-up actions are often not recorded in one structured system. This makes it difficult to monitor student progress and identify students who need additional support.

2. Proposed Solution

Develop a mentoring management platform that allows the faculty to assign mentors to students, schedule mentoring meetings, record discussion notes, track student progress, and escalate important concerns to the relevant academic or administrative officers.

3. Business Objectives

- Create a structured mentor-student allocation process.
- Maintain mentoring meeting records and follow-up actions.
- Track student progress and support requirements.
- Identify students who need academic, career, or personal development support.
- Generate mentoring reports for faculty review.

4. Key Stakeholders

Mentors / academic staff, Students, Faculty coordinator, Student support unit, Faculty management.

5. MVP Scope for 2 Months

Included in MVP	Not Included in MVP / Future Scope
Core web application with login and user roles	Advanced AI analytics
Main data entry and management screens	Full mobile app unless the team has sufficient capacity
Dashboard and basic reports	Integration with university ERP/LMS
Basic notifications by email or system alert	Payment gateway or complex third-party integrations
Testing, documentation, and deployment guide	Enterprise-level security audit

6. Key Functional Requirements

- Admin login and role management
- Mentor profile and student profile management
- Mentor-student allocation
- Meeting scheduling and reminders
- Mentoring session notes
- Student progress status
- Issue category and escalation workflow
- Dashboard: active mentors, allocated students, completed meetings, pending follow-ups
- Reports by mentor, batch, degree, and risk status

7. Non-Functional Requirements

- Responsive user interface suitable for laptop and mobile browser use.
- Secure login and role-based access control.
- Input validation and proper error messages.
- Basic audit trail for important updates.
- Database backup and restore instructions.
- Usability-focused screens with simple navigation.

8. Sample User Stories

User Role	User Story
Admin	As an admin, I want to create and manage user accounts so that system access is controlled.
Coordinator	As a coordinator, I want to view a dashboard so that I can monitor progress quickly.
Student	As a student, I want to view relevant information and submit required details online.
Management	As faculty management, I want to download reports so that decisions can be made using updated data.

9. Expected Deliverables

- Project charter and requirement specification.
- UI wireframes and final screen designs.
- Database ER diagram and schema.
- Working MVP with source code.
- Test plan, test cases, defect log, and UAT evidence.
- User manual and deployment guide.
- Final presentation and project demonstration.

10. Suggested Future Enhancements

- Student self-reflection form
- Confidential counselling referral workflow
- AI-based early warning indicators
- Calendar integration
- Mobile notification support